The density interpretation of Theorem 4 of Erdős, Graham, Ruzsa and Straus

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Abstract

For each fixed $0 < \alpha \leq 1/2$, we give the natural-density limits of the proportions of integers up to n^α which divide, or do not divide, $\binom{2n}{n}$. The constants have absolutely convergent integral series and effective truncation bounds. The nondivisor constant tends to zero as $\alpha \downarrow 0$, whereas the divisor constant tends to one. Neither proportion has a limit along all integers n in this range. We prove the uniform prime counts needed for the localized moment calculation and give certified numerical intervals. No claim is made for $1/2 < \alpha < 1$, or for an effective exceptional-set rate in n.

1 Introduction

Write $B_n = \binom{2n}{n}$. Theorem 4 on p. 89 of Erdős–Graham–Ruzsa–Straus [1] reads as follows (the mathematical notation and the final parenthesis are retained):

> THEOREM 4. *For $\alpha < 1$,*
>
> $$\left|\left\{m : 1 \leq m \leq n^\alpha,\ m \nmid \binom{2n}{n}\right\}\right| = c(\alpha)n^\alpha + o(n^\alpha),$$
>
> *where $c(\alpha) \to 1$ as $\alpha \to 0$. (In fact, $c(\alpha)$ can be explicitly calculated.)*

The qualifier "for almost all n" occurs in the preceding display (6) and its introduction, but not in the theorem itself. In the range $0 < \alpha \leq 1/2$, an interpretation as a limit along all n is false. Even after the density qualifier is supplied, the direction of divisibility in the assertion about the constant tending to one is reversed: it is the divisor proportion that has that limit.

We prove the density statement for $0 < \alpha \leq 1/2$, with an explicit series for both constants and certified intervals at four values of α. We also correct the endpoint interpretation of display (6). Nothing here is asserted for $1/2 < \alpha < 1$, although the printed theorem states the wider range $\alpha < 1$.

2 Related work

The main question of [1], whether $f(n) = \sum_{p \le n,\ p \nmid B_n} 1/p$ is bounded, is Problem 377 in Bloom's list of Erdős problems [3]; this note does not address it. Several later papers determine densities of sets of *indices* n defined by divisibility properties of B_n. Pomerance [6] shows that, for each fixed $k \ge 1$, $n + k \mid B_n$ for a set of n of density one. Sanna [8] bounds the upper density of $\{n : n \mid B_n\}$. Ford and Konyagin [4] prove that $\{n : n^\ell \mid B_n\}$ has a positive density for each $\ell \ge 1$ and evaluate these densities numerically. Pomerance [7] proves density-one divisibility statements for products of consecutive integers. These results concern the density of indices n, not the proportion of divisors of B_n among $m \le n^\alpha$ for a typical n, which is the quantity of Theorem 4 of [1] and of this note. In the formal-conjectures repository, a change merged in September 2026 [5] replaced a cofinite-convergence formalization of the reciprocal-prime result of [1] by a density-one statement; that is a different statement from Theorem 4.

In a bounded search completed on 10 October 2026 (reviews and forward citations of [1], the problem page and its forum, OEIS, and public code repositories), we did not locate an earlier published correction of the printed Theorem 4, or an earlier explicit evaluation of the densities studied here. The MathSciNet review of [1] was not accessible to us.

3 Statement and constants

For real $x = n^\alpha$ put

$$N_\alpha(n) = \frac{\#\{1 \le m \le x : m \nmid B_n\}}{x}, \qquad D_\alpha(n) = \frac{\#\{1 \le m \le x : m \mid B_n\}}{x}.$$

Thus $N_\alpha(n) + D_\alpha(n) = \lfloor x \rfloor / x$, rather than exactly one. Convergence in natural density to c means that, for every $\eta > 0$, $X^{-1}\#\{n \le X : |u(n) - c| > \eta\} \to 0$. Throughout the proof X may be restricted to positive integers; replacing a real X by $\lfloor X \rfloor$ changes normalized counts by at most $1/X$.

For $0 < t < 1$ define

$$w(t) = 2^{-\lfloor 1/t \rfloor}, \qquad g(t) = \frac{w(t)}{t}, \qquad A(y) = \int_0^y g(t)\,dt \quad (0 < y \le 1).$$

Values at the countably many breakpoints do not affect integrals. Set

$$J_0(\alpha) = 1, \qquad J_k(\alpha) = \frac{1}{k!} \int_{\substack{t_i > 0 \ (1 \le i \le k) \\ t_1 + \cdots + t_k \le \alpha}} \prod_{i=1}^{k} g(t_i)\,dt_1 \cdots dt_k \quad (k \ge 1).$$

Theorem 3.1 (Corrected Theorem 4, for $0 < \alpha \le 1/2$)**.** *For every fixed $0 < \alpha \le 1/2$ and every $\eta > 0$,*

$$\lim_{X \to \infty} \frac{1}{X} \#\{n \le X : |N_\alpha(n) - c_{\mathrm{N}}(\alpha)| > \eta\} = 0,$$

$$\lim_{X\to\infty}\frac{1}{X}\#\{n\le X : |D_\alpha(n)-c_{\mathrm{D}}(\alpha)|>\eta\}=0,$$

where the absolutely convergent series are

$$\begin{aligned} c_{\mathrm{D}}(\alpha)&=\sum_{k=0}^{\infty}(-1)^k J_k(\alpha),\\ c_{\mathrm{N}}(\alpha)&=\sum_{k=1}^{\infty}(-1)^{k+1}J_k(\alpha)=1-c_{\mathrm{D}}(\alpha). \end{aligned} \tag{DENSITY}$$

If $s=\lfloor 1/\alpha\rfloor\ge 2$*, then*

$$A(\alpha)=2^{-s}\log((s+1)\alpha)+\sum_{r=s+1}^{\infty}2^{-r}\log(1+1/r), \tag{1}$$

$$0<A(\alpha)\le A_s:=\frac{2^{1-s}}{s}\le\frac{1}{4},\qquad 0\le J_k(\alpha)\le\frac{A(\alpha)^k}{k!}, \tag{2}$$

$$A(\alpha)(1-A(\alpha)/2)\le c_{\mathrm{N}}(\alpha)\le A(\alpha)<\frac{1}{4}. \tag{3}$$

For every integer $K\ge 0$*, writing* $E_K(u)=u^{K+1}/((K+1)!(1-u/(K+2)))$*,*

$$\left|c_{\mathrm{D}}(\alpha)-\sum_{k=0}^{K}(-1)^k J_k(\alpha)\right|\le E_K(A(\alpha))\le E_K(A_s). \tag{TAIL}$$

The same bound holds for the nondivisor series truncated at $k=K$ *(the empty sum is zero). Both proportions also converge in Cesàro mean.*

Proposition 3.2. *For every fixed* $0<\alpha\le 1/2$*, neither* $N_\alpha(n)$ *nor* $D_\alpha(n)$ *has a limit as* $n\to\infty$ *through all integers. For every integer* $h\ge 0$*,*

$$v_2\left(\binom{2^{h+1}}{2^h}\right)=1,\qquad N_\alpha(2^h)\ge\frac{\lfloor 2^{h\alpha}/4\rfloor}{2^{h\alpha}}\ge\frac{1}{4}-2^{-h\alpha}.$$

Moreover $c_{\mathrm{N}}(\alpha)\to 0$ *and* $c_{\mathrm{D}}(\alpha)\to 1$ *as* $\alpha\downarrow 0$*.*

Remark 3.3 (Effectiveness and scope). The constant can be enclosed to any prescribed rational accuracy when α is rational, and for a computable real α when an approximation procedure for α is supplied. This is a claim about the construction in Section 6; the supplied program implements $\alpha=1/s$ for integers $s\ge 2$. All numerical error bounds below are explicit. We give no effective onset or rate for the exceptional sets in n. The quantifiers in Theorem 3.1 are for each fixed α and η, not uniformly in them. Nothing is claimed for $1/2<\alpha<1$.

Lemma 3.4 (Integral bounds and alternating-series bracket)**.** *The formulas and inequalities* (1)–(TAIL) *hold for the constants defined by* (DENSITY)*.*

Proof. Integrate $2^{-r}/t$ on each interval $(1/(r+1), 1/r)$ to obtain (1). Since $\log(1+1/r) \le 1/r$ and $\alpha \le 1/s$,

$$A(\alpha) \le \sum_{r \ge s} \frac{2^{-r}}{r} \le \frac{1}{s} \sum_{r \ge s} 2^{-r} = A_s.$$

Dropping the simplex constraint gives $J_k \le A(\alpha)^k/k!$. More particularly, integrate out the last coordinate and then enlarge its permitted interval to $(0, \alpha)$ to obtain

$$(k+1)!J_{k+1}(\alpha) \le k!J_k(\alpha)A(\alpha), \qquad J_{k+1}(\alpha) \le \frac{A(\alpha)}{k+1} J_k(\alpha). \tag{4}$$

Thus the nonnegative terms decrease to zero, and the alternating-series bracket gives $J_1 - J_2 \le c_{\mathrm{N}} \le J_1$. Since $J_1 = A$ and $J_2 \le A^2/2$, this proves the non-strict parts of (3). The strict inequality follows from

$$A(\alpha) \le A(1/2) = \sum_{r \ge 2} 2^{-r} \log(1 + 1/r) < \frac{1}{2} \sum_{r \ge 2} 2^{-r} = \frac{1}{4}.$$

Finally, for $j \ge 0$ and $0 \le u \le 1/4$,

$$\frac{u^{K+1+j}}{(K+1+j)!} \le \frac{u^{K+1}}{(K+1)!} \left(\frac{u}{K+2} \right)^j.$$

The geometric sum bounds the absolute tail. Its terms increase with u, so replacing $A(\alpha)$ by A_s is permitted. □

4 Proof of the density theorem

4.1 Sieve reduction

For $0 < \delta < \alpha$, let $\mathcal{P}_\delta(n)$ be the primes $n^\delta < p \le n^\alpha$ with $p \nmid B_n$. Let $F_\delta(n)$ be $n^{-\alpha}$ times the number of integers $1 \le m \le n^\alpha$ having no prime factor in $\mathcal{P}_\delta(n)$, and put

$$U_\delta(n) = \frac{\lfloor n^\alpha \rfloor}{n^\alpha} - F_\delta(n), \quad G_\delta(n) = \sum_{\substack{p \le n^\delta \\ p \nmid B_n}} \frac{1}{p}, \quad T(n) = \sum_{\substack{p \le n^\alpha \\ v_p(B_n) \ge 1}} p^{-v_p(B_n)-1}.$$

Every integer counted in U_δ is a nondivisor. Conversely, if $m \nmid B_n$, some prime satisfies $v_p(m) > v_p(B_n)$. Either $p \nmid B_n$, or $p^{v_p(B_n)+1} \mid m$ with $v_p(B_n) \ge 1$. The fraction of multiples of any positive integer d is at most $1/d$. The union bound therefore proves the pointwise sandwich

$$U_\delta(n) \le N_\alpha(n) \le U_\delta(n) + G_\delta(n) + T(n). \tag{5}$$

4.2 The localized moment calculation

We distinguish the printed counts from the uniform bounds proved below. Write

$$A_1(p;X) = \#\{p \le n \le X : p \nmid B_n\},$$
$$A_2(p,q;X) = \#\{\max(p,q) \le n \le X : \gcd(pq, B_n) = 1\}.$$

Changing a strict upper endpoint in a printed count to $n \le X$ changes its normalized value by at most $1/X$. On p. 86 of [1], immediately following (4), the text says "for each p with $x^{1/(r+1)+\epsilon} < p < x^{1/r-\epsilon}$", and the last, unnumbered display is

$$A(p;x) = x/2^r + o(x). \tag{6}$$

Here the printed $A(p;x)$ counts $p \le n < x$ with $p \nmid B_n$. For the class-III calculation on p. 88, take $x^{1/(r+1)} < p \le x^{1/r}$ and $x^{1/(t+1)} < q \le x^{1/t}$, with $p \ne q$, and order the powers p^i $(1 \le i \le r)$ and q^j $(1 \le j \le t)$ as $w_1 < \cdots < w_{r+t}$. The printed condition (5) is

$$\frac{w_{i+1}}{w_i} > x^{\epsilon} \ (1 \le i < r+t), \qquad \frac{x}{w_{r+t}} > x^{\epsilon}.$$

The text then says "then we see by (1) that", followed by

$$A(p,q;x) = \frac{x}{2^{r+t}} + o(x). \tag{7}$$

Neither quoted display prints a uniformity assertion. The $o(x)$ terms here are part of the quotations; we do not use them as uniform remainders. We now prove the uniform bounds needed for the prime summations, independently of statement (1) of [1].

Lemma 4.1 (Counting on nested scales)**.** *Let X, m be positive integers and let $Q \ge 2$ be real. Suppose the positive integers $w_1 < \cdots < w_m$ satisfy*

$$w_1 \ge Q, \qquad w_{i+1}/w_i \ge Q \ (1 \le i < m), \qquad X/w_m \ge Q.$$

If $C(X; w_1, \dots, w_m)$ counts the integers $1 \le n \le X$ for which $n \bmod w_i < w_i/2$ for every i, then

$$\left|C(X; w_1, \dots, w_m) - 2^{-m}X\right| \le \frac{8mX}{Q}. \tag{8}$$

In particular $Q = X^{\epsilon}$ gives an error at most $8mX^{1-\epsilon}$, uniformly over every collection of moduli satisfying the displayed hypotheses.

Proof. Put $h_i(n) = \mathbf{1}_{\{n \bmod w_i < w_i/2\}}$ and $c_i = \lceil w_i/2 \rceil / w_i$. Each consecutive block of w_i integers contains exactly $c_i w_i$ accepted residues. Splitting any interval I of consecutive integers into complete blocks and one remainder of length less than w_i therefore proves

$$\left|\sum_{n \in I} (h_i(n) - c_i)\right| \le w_i. \tag{9}$$

For $i < m$, let $H_{i+1}(n) = \prod_{j=i+1}^{m} h_j(n)$ on $\{1, \dots, X\}$. In this interval h_j changes value at most $2X/w_j + 2$ times, since its only changes in a period occur at residues 0 and $\lceil w_j/2 \rceil$. Every change of H_{i+1} occurs at a change of one of its factors. Thus its set of ones is a disjoint union of at most

$$R_i \le 1 + \sum_{j=i+1}^{m} (2X/w_j + 2) \le 1 + 3X \sum_{j=i+1}^{m} \frac{1}{w_j} \le 1 + \frac{6X}{w_{i+1}} \le \frac{7X}{w_{i+1}}$$

intervals of consecutive integers. The second inequality uses $X/w_j \ge Q \ge 2$, the third uses $\sum_{j=i+1}^{m} 1/w_j \le (1/w_{i+1}) \sum_{k \ge 0} Q^{-k} \le 2/w_{i+1}$, and the last uses $X/w_{i+1} \ge 1$. Apply (9) on each such interval to get

$$\left| \sum_{n=1}^{X} (h_i(n) - c_i) H_{i+1}(n) \right| \le R_i w_i \le 7X/Q \qquad (i < m).$$

For $i = m$ the same sum with the empty product equal to one has absolute value at most $w_m \le X/Q$. The exact identity

$$\prod_{i=1}^{m} h_i - \prod_{i=1}^{m} c_i = \sum_{i=1}^{m} \left(\prod_{j<i} c_j \right) (h_i - c_i) \left(\prod_{j>i} h_j \right)$$

and $0 \le c_i \le 1$ now give $|C - X \prod_i c_i| \le (7(m-1)+1)X/Q \le 7mX/Q$. Finally, $0 \le c_i - 1/2 \le 1/(2w_i)$, so another telescoping identity gives

$$0 \le \prod_i c_i - 2^{-m} \le \sum_i (c_i - 1/2) \le \frac{m}{2Q}.$$

The triangle inequality proves (8), since $7 + 1/2 < 8$. □

Application to the prime counts. Fix $0 < a < b < 1$ and $0 < \epsilon < a/4$, and suppose that the integer $X \ge 2^{1/\epsilon}$. Put $Q = X^{\epsilon}$ and $L = \lceil 2/a \rceil$. By the factorial valuation identity (28),

$$p \nmid B_n \quad \Longleftrightarrow \quad n \bmod p^i < p^i/2 \quad \text{for every } i \ge 1.$$

Indeed its ith summand is one precisely when $n \bmod p^i \ge p^i/2$, and all its summands are nonnegative. For an integer $1 \le r \le L$, if $X^a < p \le X^b$ and $X^{1/(r+1)+\epsilon} < p < X^{1/r-\epsilon}$, then $p^{r+1} > XQ^{r+1} \ge 4X$. For $n \le X$, all conditions with $i \ge r+1$ are therefore automatic. The remaining moduli are $p, p^2, \dots, p^r$; their successive ratios and their smallest modulus exceed Q, and $X/p^r > X^{r\epsilon} \ge Q$. Lemma 4.1, followed by deletion of the fewer than p integers with $n < p$, proves the explicit uniform estimate

$$\left| A_1(p; X)/X - 2^{-r} \right| \le 8rX^{-\epsilon} + p/X \le 8LX^{-\epsilon} + X^{b-1}. \tag{10}$$

For two distinct primes p, q in these trimmed bands with indices $r, t \le L$, assume also the scale inequalities of (5), with $x = X$. All powers beyond p^r and q^t

again give automatic conditions. The merged list has $m = r + t \le 2L$ distinct moduli: equality between a positive power of p and one of q would contradict unique factorization. Its smallest modulus exceeds Q, and (5) supplies the other hypotheses of Lemma 4.1. Deleting the fewer than $\max(p, q)$ initial integers gives

$$\left| A_2(p, q; X)/X - 2^{-(r+t)} \right| \le 8(r+t)X^{-\epsilon} + \frac{\max(p,q)}{X} \le 16LX^{-\epsilon} + X^{b-1}. \quad (11)$$

These estimates prove, with constants independent of the retained primes, the uniform forms required below.

For the mean of the full sum we use Theorem 2, pp. 86–87:

$$\begin{aligned} f(n) := \sum_{\substack{p \le n \\ p \nmid B_n}} \frac{1}{p}, \qquad \frac{1}{X} \sum_{n \le X} f(n) \longrightarrow c_0, \\ c_0 = \sum_{r \ge 1} 2^{-r} \log(1 + 1/r) = A(1). \end{aligned} \quad (12)$$

This full first moment is the only moment estimate imported from that paper; the uniform prime counts were proved above. In particular, we do not infer concentration of a sub-sum from the second moment of the full sum, and do not import display (6) as written.

Here and below, the errors in (10) and (11) have the following precise meaning. For fixed $0 < a < b < 1$ and $0 < \epsilon < a/4$, let $L = \lceil 2/a \rceil$ and let $e_1(X; a, b, \epsilon)$ be the maximum of $|A_1(p; X)/X - 2^{-r}|$ over $X^a < p \le X^b$, $1 \le r \le L$, in the indicated trimmed bands. Define e_2 as the analogous maximum for (11), with $r, t \le L$; an empty maximum is zero. The just-proved estimates give, for every integer $X \ge 2^{1/\epsilon}$,

$$e_1 \le 8LX^{-\epsilon} + X^{b-1}, \qquad e_2 \le 16LX^{-\epsilon} + X^{b-1}.$$

Thus both errors tend to zero with the parameters fixed. For example, both are less than any given $z > 0$ whenever

$$X > \max\left\{ 2^{1/\epsilon}, (32L/z)^{1/\epsilon}, (2/z)^{1/(1-b)} \right\}.$$

The only qualitative remainder retained here is the absolute error in (12): for every $z > 0$ it is less than z for all sufficiently large X. No numerical onset for that imported first moment, or effective exceptional-set rate for the final theorem, is claimed.

We also use the reciprocal-prime bounds of Rosser–Schoenfeld [2]. If $M(y) = \sum_{p \le y} 1/p$ and B is the prime Mertens constant, then

$$\begin{aligned} \log\log y + B - \frac{1}{2\log^2 y} < M(y) \quad (y > 1), \\ M(y) < \log\log y + B + \frac{1}{2\log^2 y} \quad (y \ge 286). \end{aligned} \quad (13)$$

Consequently, for $a \le u \le v \le b$ and $X \ge 286^{1/a}$, an interval with exponent endpoints u, v has reciprocal-prime mass differing from $\log(v/u)$ by at most

$$E_X(a) := \frac{1}{a^2 \log^2 X} + 2X^{-a}. \tag{14}$$

The $2X^{-a}$ permits either inclusion convention at the two endpoints.

Lemma 4.2 (Localized moment statement)**.** *For every fixed $0 < a < b < 1$, in natural density,*

$$S_{a,b}(n) := \sum_{\substack{n^a < p \le n^b \\ p \nmid B_n}} \frac{1}{p} \longrightarrow \lambda_{a,b} := \int_a^b g(t)\, dt. \tag{LOCAL}$$

Proof. First choose n uniformly from $\{1, \dots, X\}$ and use fixed endpoints:

$$S^X(n) = \sum_{\substack{X^a < p \le X^b \\ p \nmid B_n}} \frac{1}{p}.$$

Abbreviate $E = E_X(a)$, $H = \log(b/a) + E$, $\lambda = \lambda_{a,b}$, and $L = \lceil 2/a \rceil$. Fix $0 < \epsilon < a/4$. Remove primes whose exponent $\log p / \log X$ lies within distance ϵ of any $1/j$, $1 \le j \le L$. The union of removed intervals in $[a, b]$ has length at most $2L\epsilon$ and at most L components. Its reciprocal mass is at most

$$R_X := 2L\epsilon/a + LE, \tag{15}$$

by (14) and $\log(v/u) \le (v-u)/a$. Its limiting g-mass is at most $2L\epsilon/a$, since $w \le 1$. All other primes are in the trimmed bands; every band met has index at most L.

First moment. The count for $1 \le n \le X$ differs from $A_1(p; X)$ by at most $p \le X^b$. On the retained primes apply (10); their total reciprocal mass is at most H. There are at most L band pieces, so (14) gives

$$|\mathbb{E}S^X - \lambda| \le m_X := (e_1 + X^{b-1})H + 2LE + 4L\epsilon/a. \tag{16}$$

Here one LE bounds integration over the retained band pieces, and the other $LE + 4L\epsilon/a$ bounds the sum of the removed prime mass and its limiting integral. In particular, first letting $X \to \infty$ and then $\epsilon \downarrow 0$ proves $\mathbb{E}S^X \to \lambda$.

Excluded pairs. Choose an integer $s > 1/a$. All the primes are larger than $X^{1/s}$, so classes I and II of pp. 87–88 are empty. Pairs containing a removed prime have reciprocal mass at most $2HR_X$. For a retained prime in band r, $p^r < X^{1-r\epsilon} \le X^{1-\epsilon}$. Consecutive powers of the same prime have ratio $p > X^a > X^\epsilon$. Thus failure of condition (5) for two distinct retained primes requires

$$X^{-\epsilon} \le p^u/q^v \le X^\epsilon \quad \text{for some } 1 \le u, v \le L. \tag{17}$$

This is the ratio intended in the last excluded-pair display on p. 88, which prints $p^u q^v$. The product with two positive exponents cannot express proximity of two scales. For fixed p, u, v, the possible exponents of q lie in an interval of length at most $2\epsilon/v \le 2\epsilon$. After intersecting with $[a, b]$, (14) bounds their reciprocal mass by $2\epsilon/a + E$. Summing over p, u, v gives the explicit bound

$$\sum_{\text{excluded ordered pairs}} \frac{1}{pq} \le 2HR_X + L^2 H(2\epsilon/a + E). \tag{18}$$

This bound overcounts overlaps, which is harmless. It tends to a quantity at most $(4L + 2L^2)\log(b/a)\epsilon/a$.

Second moment and Chebyshev. The diagonal contributes at most

$$\sum_{p > X^a} \frac{1}{p^2} \le \frac{1}{X^a - 1}.$$

For the other, retained pairs satisfying (5), apply (11). Changing $n \ge \max(p, q)$ to $n \ge 1$ costs at most X^{b-1} in each normalized count. Also, summing $w(\log p/\log X)/p$ band by band differs from λ by at most LE. Hence

$$\begin{aligned} \mathbb{E}(S^X)^2 &\le (\lambda + LE)^2 + (e_2 + X^{b-1})H^2 + 2HR_X \\ &\qquad + L^2 H(2\epsilon/a + E) + \frac{1}{X^a - 1}, \end{aligned} \tag{19}$$

$$\begin{aligned} \mathbb{E}(S^X - \lambda)^2 &\le (\lambda + LE)^2 - \lambda^2 + (e_2 + X^{b-1})H^2 + 2HR_X \\ &\qquad + L^2 H(2\epsilon/a + E) + \frac{1}{X^a - 1} + 2\lambda m_X =: V_X. \end{aligned} \tag{20}$$

We used $-2\lambda \mathbb{E}S^X \le -2\lambda^2 + 2\lambda m_X$. Since $\lambda \le \log(b/a)$,

$$\limsup_{X \to \infty} V_X \le (12L + 2L^2)\log(b/a)\epsilon/a.$$

Chebyshev's inequality gives $\mathbb{P}(|S^X - \lambda| > \eta) \le V_X/\eta^2$. Letting $\epsilon \downarrow 0$ proves concentration of S^X.

Moving the endpoints. Fix $0 < \rho < 1$. For $X^{1-\rho} \le n \le X$, the symmetric difference of the two prime ranges is contained in

$$(X^{a(1-\rho)}, X^a] \ \cup \ (X^{b(1-\rho)}, X^b].$$

Consequently, for $X \ge 286^{1/(a(1-\rho))}$,

$$|S_{a,b}(n) - S^X(n)| \le 2\log\frac{1}{1-\rho} + 2E_X(a(1-\rho)). \tag{21}$$

Given $\eta > 0$, choose ρ so the logarithmic term is less than $\eta/4$ and then X so the error term is less than $\eta/4$. There are at most $X^{1-\rho}$ discarded integers. Thus the density of $|S_{a,b} - \lambda| > \eta$ is bounded by $X^{-\rho} + \mathbb{P}(|S^X - \lambda| > \eta/2)$, which tends to zero. □

Remark 4.3 (The endpoints in printed display (6)). For arbitrary endpoints, whole digit bands cannot replace their actual intersections with the interval. For example, take $a = 3/10$, $b = 9/20$. The sum in printed (6), indexed by $1/b \le k \le 1/a$, includes only $k = 3$ and gives $2^{-3}\log(4/3)$. In contrast, (LOCAL) gives

$$2^{-3}\log\frac{1/3}{3/10} + 2^{-2}\log\frac{9/20}{1/3} = 2^{-3}\log(10/9) + 2^{-2}\log(27/20).$$

The two numbers differ: eight times their difference is $\log((10/9)(27/20)^2/(4/3)) = \log(243/160) > 0$. This is an endpoint-convention defect, independent of the missing density qualifier in the printed theorem.

4.3 The finite sieve and its limit

For $n > 1$ define the finite measure

$$\mu_n = \sum_{p\in\mathcal{P}_\delta(n)} \frac{1}{p}\delta_{\log p/\log n}, \qquad \nu_\delta(dt) = \mathbf{1}_{[\delta,\alpha]}(t)g(t)\,dt.$$

The notation δ_u denotes a unit point mass at u. For $n \ge 286^{1/\alpha}$, (13) gives

$$\mu_n([\delta,\alpha]) \le \log(\alpha/\delta) + \frac{\alpha^{-2}+\delta^{-2}}{2\log^2 n}. \tag{22}$$

In particular, with $C = 1+\log(\alpha/\delta)$, this mass is at most C for $n \ge \max(286^{1/\alpha}, \exp\sqrt{(\alpha^{-2}+\delta^{-2})/2})$. The same bound holds for the mass of ν_δ.

For completeness, we explain the measure passage without an independence assumption. Partition $[\delta,\alpha]$ into m half-open intervals of length at most h. By (LOCAL), outside a set of density zero each of their m masses is within any prescribed $\zeta > 0$ of its ν_δ-mass. The sum of absolute errors of the bin masses is at most $m\zeta$. The tensor telescoping identity bounds the error of any union of bin boxes in the kth product measure by $kC^{k-1}m\zeta$, using a larger C if necessary for finitely many initial n.

On $(0,1/2]$, $g(t) \le 3/4$: on band $r \ge 2$ it is at most $(r+1)2^{-r} \le 3/4$. A slab of width ℓ in the sum of the k coordinates has $\nu_\delta^{\otimes k}$-mass at most

$$\frac{3}{4}\ell A(\alpha)^{k-1}. \tag{23}$$

Indeed fix the first $k-1$ coordinates and integrate the last over an interval of length at most ℓ. Inner and outer bin approximations to a simplex differ in a slab of width at most $2kh$. First choose h, then ζ, and then use the finite union of the exceptional sets for the bins. This proves, in natural density, convergence of $\mu_n^{\otimes k}$ on every simplex and every such slab to the corresponding $\nu_\delta^{\otimes k}$ mass. In particular no joint independence of the prime indicators is assumed in this step.

Set $K_\delta = \lfloor \alpha/\delta \rfloor$. Inclusion–exclusion gives the exact finite identity

$$F_\delta(n) = \sum_{k=0}^{K_\delta} (-1)^k \sum_{\substack{p_1 < \cdots < p_k \in \mathcal{P}_\delta(n) \\ d = p_1 \cdots p_k \le n^\alpha}} \frac{\lfloor n^\alpha/d \rfloor}{n^\alpha}. \tag{24}$$

The $k = 0$ term has $d = 1$. For $0 < \epsilon < \alpha$ and $d \le n^{\alpha-\epsilon}$, the error in replacing the floor term by $1/d$ is at most $n^{-\epsilon}/d$. Summed over all such squarefree products, it is at most

$$n^{-\epsilon} \prod_{p \in \mathcal{P}_\delta(n)} (1 + 1/p) \le n^{-\epsilon} \exp(\mu_n([\delta, \alpha])) \le n^{-\epsilon} e^C. \tag{25}$$

For the remaining products the error is at most $1/d$. Their total is bounded by $\sum_{k=1}^{K_\delta} \mu_n^{\otimes k}\{\alpha - \epsilon < \sum t_i \le \alpha\}/k!$. Its limit in density is at most $(3/4)\epsilon \sum_{k=1}^{K_\delta} A(\alpha)^{k-1}/k!$, by (23). This tends to zero with ϵ.

The repeated-prime part of the ordered k-tuple mass, for $k \ge 2$, is at most

$$\binom{k}{2} \left(\sum_{p > n^\delta} p^{-2} \right) C^{k-2} \le \binom{k}{2} \frac{C^{k-2}}{n^\delta - 1}. \tag{26}$$

This follows by specifying one equal pair of coordinates and dropping all other restrictions. Dividing the remaining distinct ordered sums by $k!$ converts them into the unordered sums in (24). Equations (25)–(26) and the simplex convergence prove

$$F_\delta(n) \longrightarrow c_{\mathrm{D},\delta} := \sum_{k=0}^{K_\delta} (-1)^k J_k^{(\delta)}(\alpha) \quad \text{in natural density,} \tag{27}$$

where $J_k^{(\delta)}$ restricts all coordinates to $[\delta, \alpha]$. Terms with $k > K_\delta$ are zero; a possible boundary simplex has measure zero. It follows that $U_\delta \to 1 - c_{\mathrm{D},\delta}$.

4.4 Insufficient prime powers

The elementary valuation formula

$$v_p(B_n) = \sum_{i \ge 1} \left(\left\lfloor \frac{2n}{p^i} \right\rfloor - 2 \left\lfloor \frac{n}{p^i} \right\rfloor \right) \tag{28}$$

follows by counting multiples of p^i in factorials. Its summands are zero or one; it is also the number of carries in adding $n + n$ in base p. Each digit of n at least $\lceil p/2 \rceil$ forces a carry. Fix p and a nonnegative integer V. Write $a_p = \lceil p/2 \rceil$, $b_p = \lfloor p/2 \rfloor$, and $h_p = \lfloor \log_p X \rfloor$. A block of p^{h_p} consecutive integers has exactly

$$Q_{p,V}(h_p) = \sum_{j=0}^{\min(V, h_p)} \binom{h_p}{j} b_p^j a_p^{h_p - j}$$

residues with at most V large digits among the lowest h_p digits. There are at most $\lceil X/p^{h_p} \rceil \le 2X/p^{h_p}$ blocks, including a partial one. Hence

$$\frac{1}{X}\#\{n \le X : v_p(B_n) \le V\} \le \frac{2Q_{p,V}(h_p)}{p^{h_p}}. \tag{29}$$

For $h_p \ge \max(V, 1)$ the right side is at most $2(V+1)h_p^V (a_p/p)^{h_p}$, since $b_p \le a_p$ and $\binom{h_p}{j} \le h_p^V$. It tends to zero, including when $p = 2$.

For an integer $H \ge 2$, the part of $T(n)$ from $p > H$ is at most $\sum_{m>H} m^{-2} \le 1/H$, by integration of t^{-2}. If all $p \le H$ have $v_p(B_n) > V$, the other part is at most $H2^{-V-2}$. Given $\eta > 0$, choose H and V so that $1/H + H2^{-V-2} < \eta$. Then

$$\frac{1}{X}\#\{n \le X : T(n) > \eta\} \le \sum_{p \le H} \frac{2Q_{p,V}(h_p)}{p^{h_p}} \longrightarrow 0.$$

Thus $T \to 0$ in natural density with every discarded bound specified.

4.5 Small primes and assembly

Fix $\rho > 0$ with $\delta < 1 - \rho$. Pointwise, $f(n) \ge G_\delta(n) + S_{\delta,1-\rho}(n)$. The local sum is bounded for all sufficiently large n by (13). A bounded sequence converging in density converges in mean: if $|u(n)| \le C$ and $u(n) \to u$ in density, the mean of $|u(n) - u|$ is at most η plus $C + |u|$ times the proportion where $|u(n) - u| > \eta$; finitely many initial exceptions contribute their fixed total divided by X. Apply this fact to (LOCAL) and then use (12) to obtain

$$\limsup_{X\to\infty} \frac{1}{X} \sum_{n \le X} G_\delta(n) \le c_0 - \int_\delta^{1-\rho} g(t)\, dt.$$

Let $\rho \downarrow 0$. Monotone convergence gives

$$\begin{aligned} \limsup_{X\to\infty} \frac{1}{X} \sum_{n \le X} G_\delta(n) &\le A(\delta), \\ \limsup_{X\to\infty} \frac{1}{X}\#\{n \le X : G_\delta(n) > \eta\} &\le \frac{A(\delta)}{\eta}. \end{aligned} \tag{30}$$

The second inequality is Markov's inequality. The full second moment and a uniform-integrability argument are unnecessary here.

Deleting the measure on $(0, \delta)$ changes its total mass by $A(\delta)$. Dropping the simplex restriction, or telescoping product measures, gives for $k \ge 1$

$$0 \le J_k - J_k^{(\delta)} \le \frac{A(\alpha)^k - (A(\alpha) - A(\delta))^k}{k!} \le \frac{A(\delta)A(\alpha)^{k-1}}{(k-1)!}.$$

Consequently

$$|c_{\mathrm{D},\delta} - c_{\mathrm{D}}(\alpha)| \le A(\delta)e^{A(\alpha)} < 2A(\delta). \tag{31}$$

Given $\eta > 0$, choose any sufficiently small $\delta > 0$ so that $2A(\delta) < \eta/4$. Outside the union of

$$\{|U_\delta - (1 - c_{\mathrm{D},\delta})| > \eta/4\},$$
$$\{T > \eta/4\}, \qquad \{G_\delta > \eta/4\},$$

the sandwich (5) implies $|N_\alpha - c_{\mathrm{N}}| \le \eta$. The first two exceptional sets have density zero, and the third has upper density at most $4A(\delta)/\eta$. Letting $\delta \downarrow 0$ proves the first assertion of Theorem 3.1. The identity $D_\alpha + N_\alpha = \lfloor n^\alpha \rfloor / n^\alpha$, whose error from one is at most $n^{-\alpha}$, proves the second. Both sequences lie in $[0, 1]$, so the preceding bounded-convergence-in-density argument proves Cesàro convergence as well.

5 Failure of the printed statement as written

Proof of Proposition 3.2. For $n = 2^h$, all the terms in (28) with $p = 2$ vanish except $i = h + 1$, which equals one. Thus $4 \nmid B_{2^h}$ and every multiple of four is a nondivisor. The proposition's explicit lower bound follows from $\lfloor y \rfloor \ge y - 1$.

By Lemma 3.4, $c_{\mathrm{N}}(\alpha) < 1/4$ for every $0 < \alpha \le 1/2$. A limit along all integers would also be a limit in natural density, and such a limit is unique: two different limits have disjoint neighborhoods of radius less than half their separation, whereas each neighborhood would have density one. The subsequence $n = 2^h$ therefore contradicts existence of an all-integer limit for N_α. Since $N_\alpha + D_\alpha \to 1$, it also contradicts existence of one for D_α.

As $\alpha \downarrow 0$, $s = \lfloor 1/\alpha \rfloor \to \infty$ and $A(\alpha) \le 2^{1-s}/s \to 0$. The bracket $0 \le c_{\mathrm{N}}(\alpha) \le A(\alpha)$ gives $c_{\mathrm{N}}(\alpha) \to 0$ and $c_{\mathrm{D}}(\alpha) = 1 - c_{\mathrm{N}}(\alpha) \to 1$. □

Thus the nondivisor direction of the printed small-α assertion fails even with the density qualifier restored. Switching to divisors repairs that limit, but does not produce an all-integer limit. These conclusions concern $0 < \alpha \le 1/2$; the interval $1/2 < \alpha < 1$ remains outside the statement proved here.

6 Certified evaluation

The accompanying program `code/density_intervals.py` uses only exact integers and rational numbers. We describe each error bound, including the discretization error. The stated table uses $\alpha = 1/s$ with $s \in \{2, 3, 4, 5\}$, mesh $M = 16384$, cutoff $R = 64$, $K = 8$ terms, and $Q = 2^{56}$. The program as shipped implements $\alpha = 1/s$ for integers $s \ge 2$; the construction extends to any rational $0 < \alpha \le 1/2$ by clipping the top band at α.

6.1 Rational bin masses and logarithms

Let $h = \alpha/M$ and $I_i = ((i-1)h, ih]$, $1 \le i \le M$. First restrict $\nu(dt) = g(t)\,dt$ to $[1/R, \alpha]$, with $R > s$. An intersection $[a, b]$ of a bin and a band $[1/(r+1), 1/r]$

has mass $2^{-r}\log(b/a)$. Zero-length intersections are ignored. Here $1 \le b/a \le (r+1)/r \le 3/2$. For $z = (b/a-1)/(b/a+1)$, put

$$\ell_J = 2\sum_{j=0}^{J-1} \frac{z^{2j+1}}{2j+1}.$$

Integrating the geometric series for $2/(1-z^2)$, or summing its positive terms, gives the rational enclosure

$$\ell_J \le \log(b/a) \le \ell_J + \frac{2z^{2J+1}}{(2J+1)(1-z^2)}. \tag{32}$$

Indeed each denominator in the omitted series is at least $2J+1$. Here $0 \le z \le 1/5$, so the bound tends to zero. Increase J until the displayed error is less than $1/(Q2^{20})$. Round the lower and upper bounds for $2^{-r}\log(b/a)$ down and up, respectively, to integer multiples of $1/Q$. Sum segment bounds within each bin to get

$$q_i^-/Q \le \nu(I_i \cap [1/R, \alpha]) \le q_i^+/Q.$$

Every operation defining these integers is exact.

6.2 Simplex bounds and exact convolution

For a tuple of bins, $\sum i_j \le M$ implies that its whole box lies in $\sum t_j \le \alpha$. Conversely, an intersecting box has $\sum i_j \le M+k-1$, and in particular $\sum i_j \le M+k$. Thus, with $P_\pm(z) = \sum_{i=1}^{M} q_i^\pm z^i$, lower and upper bounds for the truncated-measure J_k are

$$L_k = \frac{\sum_{j=0}^{M}[z^j]P_-(z)^k}{k!Q^k}, \qquad U_k = \frac{\sum_{j=0}^{M+k}[z^j]P_+(z)^k}{k!Q^k}. \tag{33}$$

Repeated bins are allowed: products of their masses are exactly the product-measure masses of boxes. Divide by $k!$ because J_k uses ordered integration variables with that prefactor. Alternating signs are applied outwards, using $[-U_k, -L_k]$ when k is odd.

To compute the coefficients, pack each polynomial into base 2^B, where

$$B = 8\left\lfloor \frac{56K+8}{8} \right\rfloor = 456 \quad \text{for } K = 8.$$

More generally the code replaces 56 by the chosen bit precision. It checks $\sum_i q_i^+ < Q$. Thus every coefficient and every prefix sum in any power $1 \le k \le K$ is strictly less than $Q^k \le Q^K < 2^B - 1$. There are no carries between packed coefficients. Integer multiplication and masking after degree $M+K$ therefore give exact truncated polynomial multiplication; discarded high degrees cannot affect lower ones because all degrees are nonnegative. Reduction of a selected prefix modulo 2^B-1 sums its base-2^B digits exactly: $2^B \equiv 1$ modulo 2^B-1, and the digit sum is less than that modulus. This proves the convolution algorithm, without any floating-point rounding assumption.

6.3 All remaining errors and convergence

The omitted mass below $1/R$ satisfies

$$\tau := \int_0^{1/R} g(t)\,dt = \sum_{r\ge R} 2^{-r}\log(1+1/r) \le \tau_R := \frac{2^{1-R}}{R}.$$

Writing $A = A(\alpha)$, the absolute change in the full alternating series is bounded by

$$\sum_{k\ge 1}\frac{A^k-(A-\tau)^k}{k!} = e^A - e^{A-\tau} \le \tau e^A \le \tau_R e^{A_s} < 2\tau_R.$$

Truncating the k-series adds at most $E_K(A_s)$ by (TAIL). Add

$$2\tau_R + E_K(A_s) \tag{34}$$

to each end of the interval computed from (33). For $s = 2$, $R = 64$, $K = 8$, this radius is strictly less than 0.00000000001078182876. Nondivisor bounds are one minus the reversed divisor bounds. Decimal endpoints are rounded outwards using integer division, so they remain rational enclosures.

Here is also an explicit convergence guarantee. Let $\eta_Q = \sum_i (q_i^+ - q_i^-)/Q$. There are at most $M + R$ positive bin–band intersections, since the common refinement of the two partitions has at most the sum of their numbers of pieces. For each segment the scaled logarithm enclosure has length less than 2^{-20}; after rounding its integer width is at most two. Therefore

$$\eta_Q \le \frac{2(M+R)}{Q}. \tag{35}$$

For the exact bin masses, the boxes included in the upper simplex bound but not the lower one lie in a slab of width at most $2kh$. By (23), summing their contributions divided by $k!$ is at most

$$\sum_{k\ge 1}\frac{2kh(3/4)A_s^{k-1}}{k!} = \frac{3}{2}he^{A_s} < 3h.$$

Let m_- and m_+ be the total lower and upper rounded masses. They differ by η_Q, and $m_+ \le A_s + \eta_Q$. The sum of the rounding contributions to the interval width is at most $\sum_{k\ge1}(m_+^k - m_-^k)/k! \le \eta_Q e^{m_+} < 3\eta_Q$ when $A_s + \eta_Q \le 1$. Together with (34), the unrounded enclosure width is at most

$$3h + 3\eta_Q + 4\tau_R + 2E_K(A_s). \tag{36}$$

Reporting to d decimal places adds at most $2\cdot 10^{-d}$ to this width. Increasing M, R, K, Q so that each displayed bound is below its allotted tolerance gives any prescribed rational accuracy. All choices have explicit bounds: one may test (36) using rational upper bounds $3h$, $6(M+R)/Q$, $4\tau_R$, and $2E_K(A_s)$,

then increase the parameters until their sum is small enough. The assertion $\sum q_i^+ < Q$ eventually holds since $A_s \le 1/4$ and (35) tends to zero.

For a real input α specified by computable rational approximations, a modulus also follows from the same slab estimate. If $0 < \alpha \le \beta \le 1/2$, then

$$0 \le J_k(\beta) - J_k(\alpha) \le \frac{3(\beta-\alpha)A(\beta)^{k-1}}{4k!}, \quad k \ge 1.$$

Summing shows $|c_{\mathrm{D}}(\beta) - c_{\mathrm{D}}(\alpha)|$ and $|c_{\mathrm{N}}(\beta) - c_{\mathrm{N}}(\alpha)|$ are at most $(3/4)e^{1/4}(\beta - \alpha) < 2(\beta - \alpha)$. This supplies the approximation interpretation in Remark 3.3; it asserts no algorithm for an unspecified, noncomputable real input.

6.4 Certified intervals and reproduction

The following endpoints are transcribed exactly from `code/density_16384.txt`. They are outward-rounded rational intervals, not confidence intervals.

α	$c_{\mathrm{N}}(\alpha)$: nondivisors	$c_{\mathrm{D}}(\alpha)$: divisors
1/2	[0.160401041573, 0.160401748642]	[0.839598251358, 0.839598958427]
1/3	[0.059862571395, 0.059862608229]	[0.940137391771, 0.940137428605]
1/4	[0.023932466470, 0.023932468474]	[0.976067531526, 0.976067533530]
1/5	[0.009987263535, 0.009987263648]	[0.990012736352, 0.990012736465]

For $\alpha = 1/2$ the displayed width is 0.000000707069. The mesh-4096 interval is $[0.160399449123, 0.160402279170]$, of width 0.000002830047. The ratio of these displayed widths is greater than four, since $2830047 > 4 \cdot 707069$.

Run the following from the directory containing `main.tex`; the Python programs need only the standard library:

```
python3 code/density_intervals.py --mesh 4096 \
  > density_4096.rerun.txt
python3 code/density_intervals.py --mesh 16384 \
  > density_16384.rerun.txt
cmp code/density_4096.txt density_4096.rerun.txt
cmp code/density_16384.txt density_16384.rerun.txt
python3 code/check_convolution.py
```

The convolution check compares packed powers to independent nested-loop convolution for $s \in \{2,3,4,5\}$, $M \in \{8,16,31\}$, $R = 12$, both choices of coefficient bounds, and $k = 1,2,3,4$: $4 \cdot 3 \cdot 2 \cdot 4 = 96$ exact comparisons. It prints

```
96 packed polynomial powers agree with independent nested-loop convolutions.
```

These finite checks test the implementation; the certification rests on (32)–(36). Neither the table nor the checks supply an effective exceptional-set rate for the density theorem.

Provenance and verification status

The density theorem, its proof and the certified evaluation were found by a GPT-6 Astra agent (OpenAI) on 2026-09-09; the present text, which applies the

referees' repairs, was revised by GPT-6 Astra on 2026-10-10. Two isolated cross-vendor referee reports by Claude Opus (Anthropic), dated 2026-10-10, both returned *PASS-WITH-REPAIRS* with no fatal gap. Both asked for the same three required repairs, all applied here: the decreasing-term inequality behind $0 \le c_{\mathrm{N}}(\alpha) \le A(\alpha)$ (Lemma 3.4); the localized moment statement written out (Lemma 4.2); and the scope $0 < \alpha \le 1/2$ (Remark 3.3). Referee 1 also suggested an optional simplification of the small-prime argument, applied in Section 4.5. In a second round both referees asked for a self-contained proof of the uniform forms of the two counts taken from the 1975 paper; this is Lemma 4.1, which both confirmed in a third round (*PASS*). Each referee reran the certificate program at both meshes with byte-identical output, and computed the four constants by an independent method; the converged values fell inside the certified intervals. A separate Claude verifier reproduced the certificates byte for byte and checked $v_2\big(\binom{2^{h+1}}{2^h}\big) = 1$ exactly for $1 \le h \le 300$. Empirical proportions for n between about 10^9 and 10^{24} lie roughly 9–17% below the limits, depending on α and on the range of n, which is consistent with slow convergence in n. These are sanity checks, not evidence for the theorem. No formal (Lean) verification has been carried out.

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